

$$1. \quad x[n] = u[n] \cdot A e^{-\beta n} \cos(\omega_0 n)$$

$$= u[n] \cdot A e^{-\beta n} \cdot \frac{1}{2} (e^{j\omega_0 n} + e^{-j\omega_0 n})$$

$$= \frac{A}{2} u[n] e^{-(\beta - j\omega_0)n} + \frac{A}{2} u[n] e^{-(\beta + j\omega_0)n}$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \left[\frac{A}{2} u[n] e^{-(\beta - j\omega_0)n} + \frac{A}{2} u[n] e^{-(\beta + j\omega_0)n} \right] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} \frac{A}{2} e^{-(\beta + j(\omega - \omega_0))n} + \sum_{n=0}^{\infty} \frac{A}{2} e^{-(\beta + j(\omega + \omega_0))n}$$

$$= \frac{A}{2} \left[\frac{1}{1 - e^{-(\beta + j(\omega - \omega_0))}} \right] + \frac{A}{2} \left[\frac{1}{1 - e^{-(\beta + j(\omega + \omega_0))}} \right] \quad \#$$

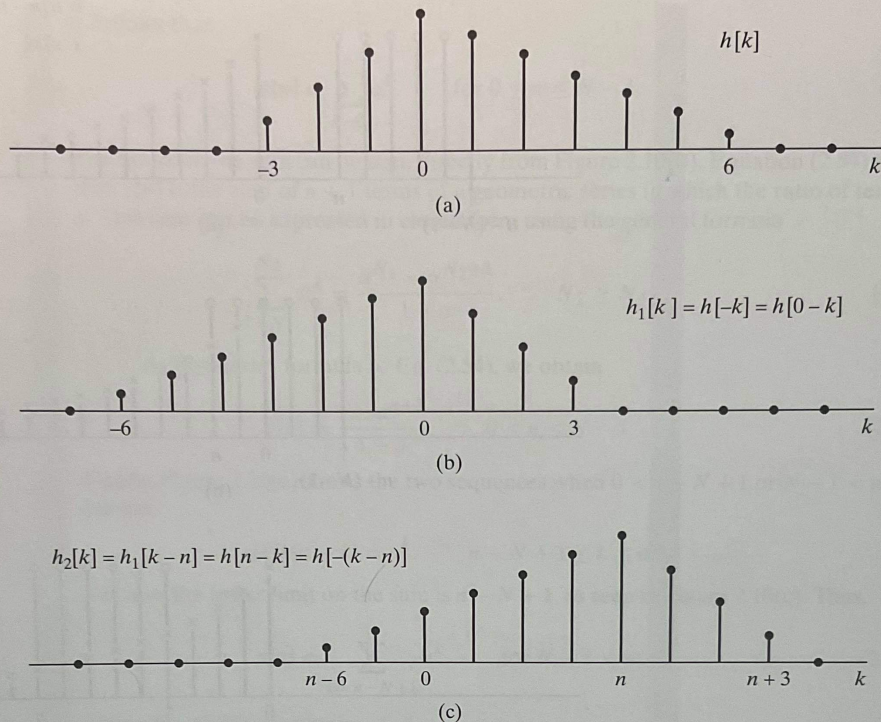


Figure 2.9 Forming the sequence $h[n-k]$. (a) The sequence $h[k]$ as a function of k . (b) The sequence $h[-k]$ as a function of k . (c) The sequence $h[n-k] = h[-(k-n)]$ as a function of k for $n=4$.

$$2. (a) y[n] = x[n] * x[-n]$$

$$= \sum_k x[k] \underbrace{x[-n+k]}_{\text{forming process like } h[n-k] \text{ in fig. 2.9}}$$

original: $x[k]$

reversal: $x_1[k] = x[k]$

delay $x_1[k]$
by n samples: $x_2[k] = x_1[k-n] = x[k-n]$

$$\therefore y[n] = \sum_k x[k] x[k] \quad \#$$

$$(b) Y(e^{j\omega}) = F\{y[n]\}$$

$$= F\{x[n]\} \cdot F\{x[-n]\}$$

$$= X(e^{j\omega}) \cdot X^*(e^{j\omega})$$

$$= |X(e^{j\omega})|^2 \quad \#$$

$$2.(c) \quad y[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} Y(e^{j\omega}) e^{j\omega n} d\omega$$

$$y[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} Y(e^{j\omega}) d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega \quad (\text{from 2.(b)})$$

$$y[0] = \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad (\text{from 2.(a)})$$

$$\Rightarrow \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega \quad \#$$

$$3.(a) \quad h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\frac{\omega}{3}} e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \frac{1}{j(n-\frac{1}{3})} e^{j(n-\frac{1}{3})\omega} \Big|_{-\pi}^{\pi}$$

$$= \frac{1}{2j} \frac{1}{(n-\frac{1}{3})\pi} \left[e^{j(n-\frac{1}{3})\pi} - e^{-j(n-\frac{1}{3})\pi} \right]$$

$$= \frac{1}{(n-\frac{1}{3})\pi} \sin\left[\left(n-\frac{1}{3}\right)\pi\right]$$

$$= \text{sinc}\left(n-\frac{1}{3}\right) \quad \#$$